

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ404			
Course Title	CODING THEORY			
Type of Course	Elective			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial	Practicum	Total Hours
		per week	per week	
	4	4	-	60
Pre-requisites	Linear Algebra, Algebra			
Course Summary	The course helps the student to understand various algebraic codes, - their encoding and decoding methods and the mathematical tools used in their design.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Construct the parity check/generator matrix of a linear code.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Calculate bounds on rate and distance of a given linear code using various bounds.	An	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Design cyclic codes of a given rate and distance parameters and decode it using various standard decoding procedures.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text	Huffman, W. Cary, and Vera Pless. Fundamentals of error-correcting codes. Cambridge university press, 2010.			
Module	Unit	Content	Hrs (48+12)	External Marks (70)
I	Linear Codes		12	
	Text Sections: 1.1, 1.2, 1.4, 1.5.1 to 1.5.3, 1.8, 1.10, 1.11.2			
	1	Binary and Prime Fields		
	2	Linear Codes - Generator and Parity Check Matrix		
	3	Weights and Distances		
	4	Punchuring, Shortening and Extension		
	5	Hamming Codes		
	6	Reed Muller Codes		
	7	Encoding Linear Codes		
II	Bounds on Linear Codes		5	
	Text Sections: 2.2, 2.4, 2.8			
	8	Plotkin Bound		
	9	Singleton Bound and MDS codes		
	10	Gilbert - Varshamov Lower Bound		
	11	Asymptotic Singleton and Plotkin Bounds		
III	Finite Fields and Cyclic Codes		15	
	Text Sections: 3.1 to 3.7 and 4.1, 4.2, 4.5.			
	12	Finite fields and elementary properties		
	13	Polynomials and Euclid's Algorithm		
	14	Primitive Elements		
	15	Construction of Finite fields		

	16	Cyclotomic Polynomials		
	17	Basic Theory of Cyclic Codes		
	18	BCH Bound.		
IV	BCH and Reed Solomon Codes Text Sections: 5.1, 5.2, 5.3, 5.4.1 to 5.4.3		16	
	18	BCH Codes		
	19	Reed Solomon Codes and their generalization.		
	20	Peterson–Gorenstein–Zierler Decoding Algorithm		
	21	Berlekamp Massey Decoding Algorithm		
	22	Sugiyama Decoding Algorithm (Euclid’s Algorithm)		
V	OPEN ENDED		12	-
	1	List decoding and Guruswami Sudan Algorithm		
	2	Weight Distributions of Codes and McWilliams Identities		
	3	Self-dual codes.		
	4	Codes on Projective Planes		
	5	Codes over Z_4		
	6	Convolutional Codes		
References	1. E. F. Assmus, Jr. and J. D. Key, Designs and Their Codes. London: Cambridge University Press, 1993. 2. R. E. Blahut, Theory and Practice of Error Control Codes. Reading, MA: Addison-Wesley, 1983.			

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	0	3	1	2	1	3	0	1
CO 2	3	2	2	0	3	1	3	1	3	0	1
CO 3	3	3	2	0	3	1	3	1	3	0	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓